

MICHAEL LEE

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EDUCATION

Colorado School of Mines — Golden, CO

Starting Aug 2027 | Expected Dec 2028

- M.S. in Mechanical Engineering (Concentration: Thermal, Fluid & Energy Systems)

Northwestern University — Evanston, IL

Jun 2025

- B.S. in Mechanical Engineering (Concentration: Aerospace Engineering)
- Courses: Propulsion, Compressible Flow, Heat Transfer, CFD, Thermal Systems Design, Mechanical Design, Manufacturing

PROFESSIONAL EXPERIENCE

Entegris — Colorado Springs, CO

Jul 2026 – Present

Metrology Engineering Co-op

- Design, fabricate, and integrate custom inspection fixtures using SolidWorks CAD and 3D printing to improve coordinate-measuring machine (CMM) measurement repeatability and accuracy
- Interpret technical drawings, GD&T, and Inspection Methodology (IMT) documents to develop, test, and optimize CMM programs for machined and injection-molded parts, enabling accurate inspection of tight-tolerance components used in advanced semiconductor packaging
- Work closely with quality inspectors and metrology technicians to improve inspection workflows by incorporating feedback from those closest to the final product
- Automate measurement data analysis using Python and Minitab to assess CMM program capability by calculating Gage R&R metrics, bias, correlation, and Cg/Cgk indices
- Support process changes through CFMEA risk assessments and Management of Change (MOC) documentation in ETQ Reliance

PROJECTS

Gear Reducer Transmission System

Mar 2025 – Jun 2025

- Designed and integrated shaft #1 in a 3-shaft, 56 kW gearbox, reducing input speed from 2400 rpm to 337 rpm by calculating gear forces and combined stresses to guide material selection and bearing sizing
- Sized a 35 mm diameter AISI 1050 steel shaft by performing fatigue and static yield failure analyses, achieving >10 million cycles with safety factors > 1.27; selected bearings and verified L10 bearing life at operating speeds
- Built complete Siemens NX CAD shaft assembly model and detailed part drawings with standard fits and tolerances, incorporating keyways and auxiliary components (end caps, gaskets, spacers) to ensure proper manufacturing and assembly

Aerospace Testbed Design and Integration

Jan 2025 – Jun 2025

- Designed a modular wing testbed for a high school summer aerospace course, defining requirements for four subsystems — avionics, wing manufacturing, wing-fuselage attachment, and launcher — enabling 30+ students to build and test custom wings
- Developed avionics subsystem using Siemens NX CAD and circuit diagrams; processed flight sensor data in Python, achieving lift and drag coefficients within 10% of predictions
- Managed 5-month project timeline using Gantt and RAM charts to assign and track responsibilities across 7-person team, resulting in on-time delivery of deliverables, including final report, bill of materials, fabrication guide, and presentation

CFD Analysis of Rayleigh-Taylor Instabilities

Feb 2025 – Mar 2025

- Simulated Rayleigh-Taylor instabilities using ANSYS Fluent Volume of Fluid model to analyze effect of Atwood number on perturbation wavelength and growth rate
- Applied Python for FFT-based post-processing analysis and validated trends against linear stability theory; reduced growth-rate error by 5.7% using Richardson Extrapolation with mesh dependency of 2.1%

SKILLS

CAD & Engineering: SolidWorks CAD, Siemens NX CAD, GD&T

Analysis & Programming: ANSYS (Fluent CFD, Mechanical FEA), NX FEA, Python (Pandas, NumPy, SymPy), Minitab, MATLAB

Prototyping: 3D Printing, Electronics Soldering & Wiring, Manual Milling, Hand & Power Tools